# Subtracting vectors



AS Level | 9702 AS Ch.1 | Chapter 1: Kinematics | Lesson 1.7 | Sections 1.7

**Goal:** Find a change in velocity, Δv = $v_2$ − $v_1$, by adding the reverse of the first vector.

velocity acceleration

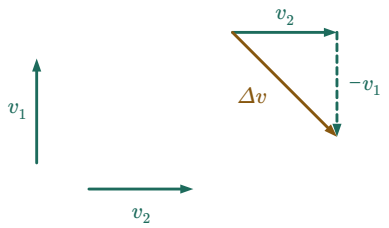
## 01 The big picture

To subtract a vector, add its reverse: $\vec{a} - \vec{b} = \vec{a} + (-\vec{b})$, where $-\vec{b}$ has the same size as $\vec{b}$ but points the opposite way.

The most common use is a change in velocity: $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$ (final minus initial). Even at constant speed, a change of direction gives a non-zero $\Delta v$.

**Change in velocity** [paraphrase]
Final velocity minus initial velocity, found as a vector.

**Fig. 1 · Δv = $v_2$ − $v_1$ = $v_2$ + (−$v_1$)**




Draw $v_2$, then add the reverse of $v_1$ at its tip. Δv runs from the start of $v_2$ to the end of −$v_1$.

## 02 Key equations

**Subtracting a vector**

$$\vec{a} - \vec{b} = \vec{a} + (-\vec{b})$$

**Change in velocity**

$$\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$$

**Symbols and units**

| | | | | | |
|---|---|---|---|---|---|
| $\vec{v}_1$, $\vec{v}_2$ | initial and final velocity | $\mathrm{m\,s^{-1}}$ | $\Delta\vec{v}$ | change in velocity | $\mathrm{m\,s^{-1}}$ |

## 03 Sample problems with solutions

**Sample 1 · A car turning a corner**
A car travelling north at 20 m s⁻¹ turns and travels east at 20 m s⁻¹. Find its change in velocity.

**Step 1.** Add the reverse of $v_1$ (20 m s⁻¹ south) to $v_2$ (20 m s⁻¹ east): the two are perpendicular.
**Step 2.** Direction: equal east and south parts, so 45° between them.

$$|\Delta v| = \sqrt{20^2 + 20^2} = 28.3\ \mathrm{m\,s^{-1}}$$

**Answer:** 28 m s⁻¹ towards the south-east (45° south of east)

## 04 Practice problems

**P1** A ball moving right at 6.0 m s⁻¹ rebounds moving left at 4.0 m s⁻¹. Find the change in velocity.
Answer: 10 m s⁻¹ to the left

**P2** A velocity changes from 3.0 m s⁻¹ east to 4.0 m s⁻¹ north. Find Δv.
Answer: 5.0 m s⁻¹ at 53° north of west

**P3** Find A − B for A = 7.0 N east and B = 2.0 N east.
Answer: 5.0 N east

**P4** A velocity changes from 10 m s⁻¹ north to 10 m s⁻¹ south. Find Δv.
Answer: 20 m s⁻¹ south

## 05 Other useful information

**Common mistakes**

✗ **Wrong:** Subtract the speeds: 20 − 20 = 0, so no change.
✓ **Right:** The direction changed, so Δv is not zero.

✗ **Wrong:** Work out $v_1$ − $v_2$.
✓ **Right:** Δv = final − initial: $v_2$ − $v_1$.

**Exam tips**

- In one dimension, give each velocity a sign (+ one way, − the other) and subtract the numbers.

**What the syllabus asks**

- 9702 topic 1: add and subtract coplanar vectors. Coursebook 1.7.

**Explore:** Practice questions for this lesson